

Restricting the Flow: Information Bottlenecks for Attribution ICLR 20'

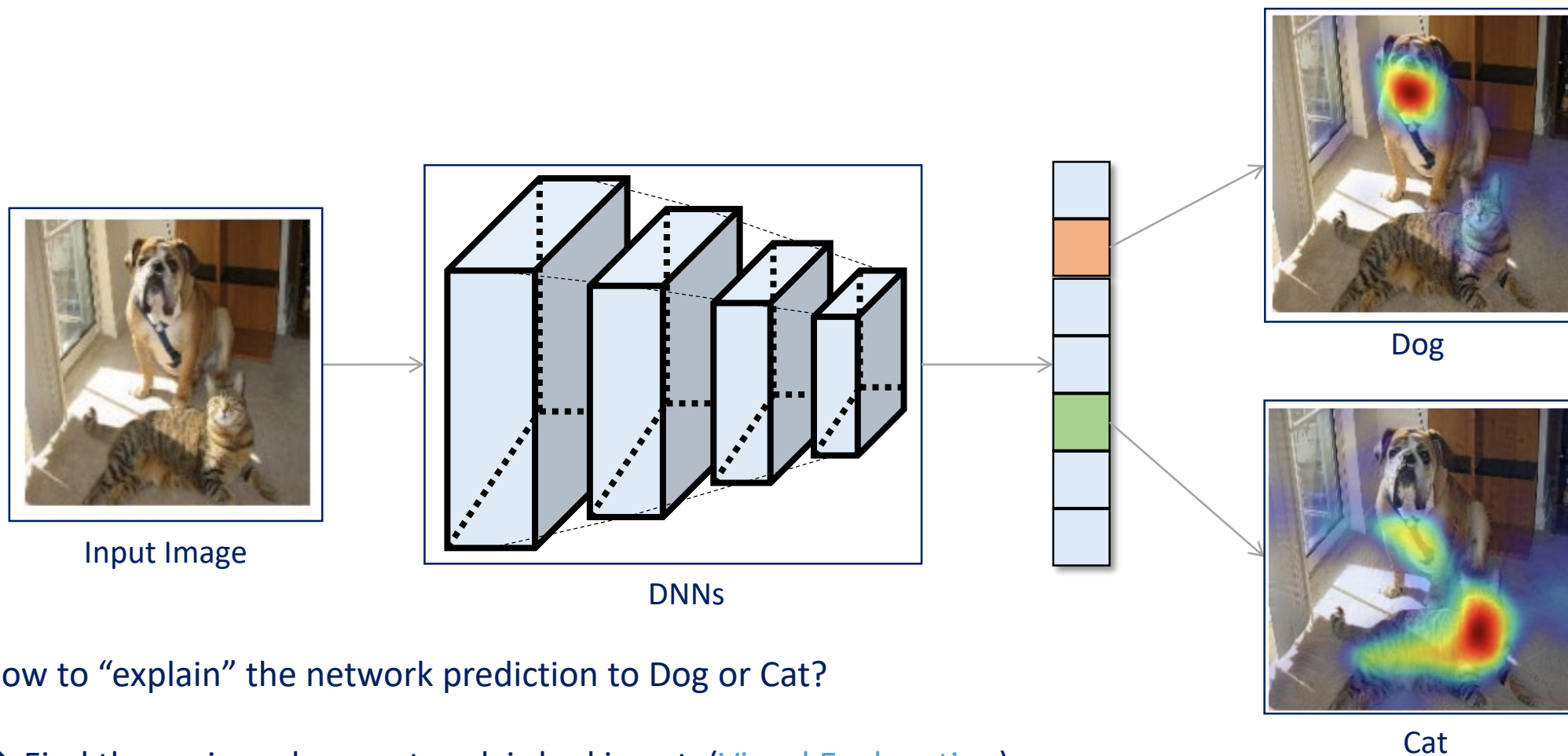
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Understanding Deep Neural Networks

- What is attribution map?: Providing insights of the DNNs' decision-making



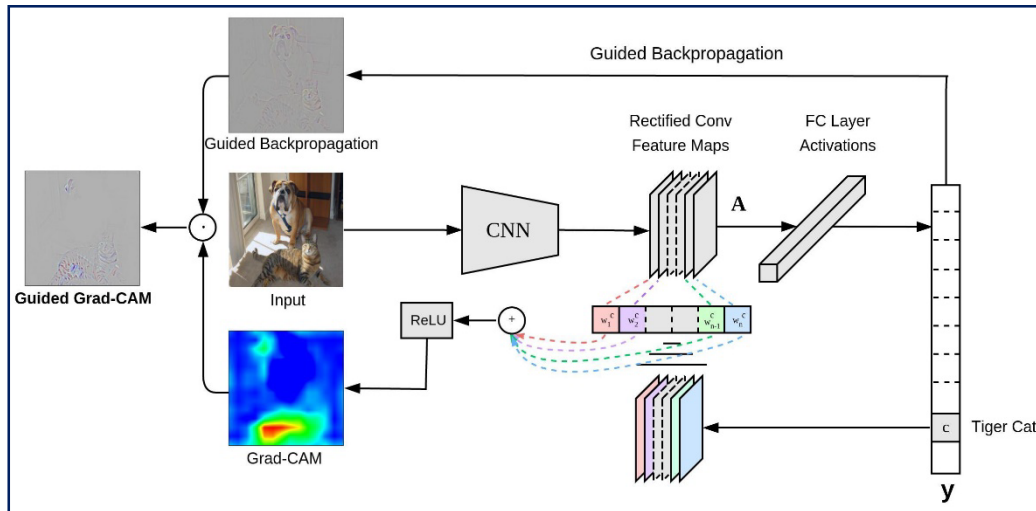
How to “explain” the network prediction to Dog or Cat?

→ Find the region where network is looking at. ([Visual Explanation](#))

Visual Explanation

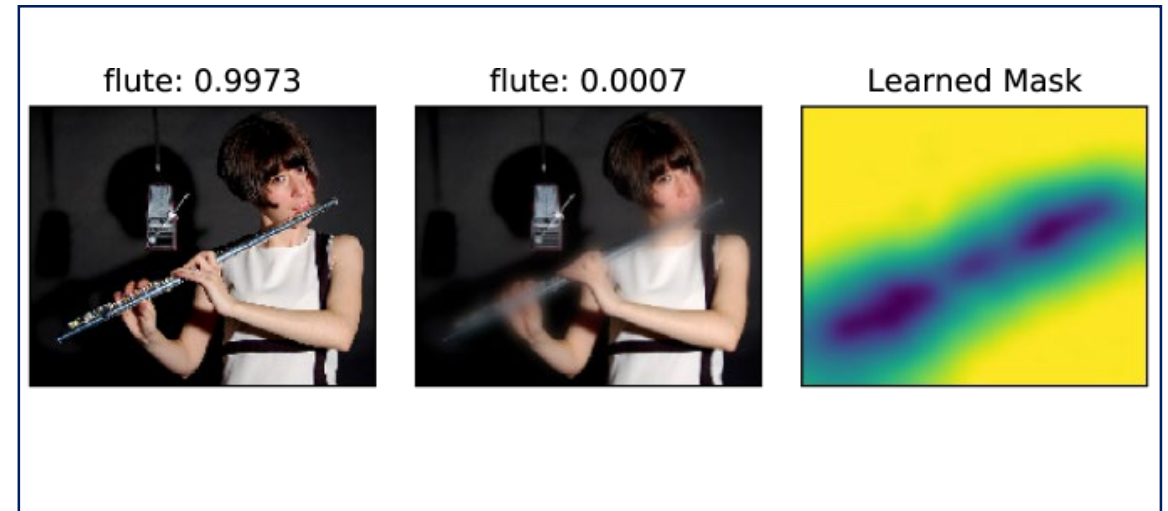
- Visualizing Attribution Maps

White-box Approach



- Pros: Simple and Fast.
- Cons: Need tractable internal components (e.g., gradient, activation), Network architecture dependency.

Black-box approach



- Pros: Model-agnostic property, Interpretability for the black-box models.
- Cons: difficult to optimize.

[Selvaraju et al. 2017; Fong et al. 2017]

Introduction

- Restricting the flow: Information Bottlenecks for Attribution (ICLR 20')

Existing attribution heatmap highlights subjectively irrelevant areas and this might correctly reflect the network's unexpected way of processing the data

→ So, the authors proposed a novel attribution method that estimates the amount of information an image region provides to the network's prediction using information bottleneck concept

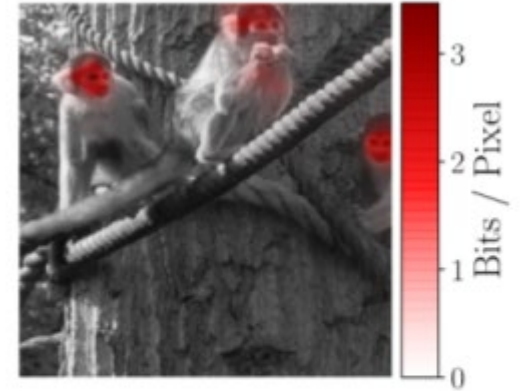


Figure 1: Exemplary heatmap of the Per-Sample Bottleneck for the VGG-16.

Preliminaries

- Information Bottleneck:

Goal: minimizing information flow + maximizing original model objective



Common way to reduce the amount of information?

$$Z = \lambda(X) \boxed{R} + (1 - \lambda(X)) \boxed{\epsilon}, \quad \lambda_i \in [0, 1]$$

$\boxed{R}$ has a blue arrow pointing down to $R = f_i(X) \rightarrow l\text{th layer output}$.
 $\boxed{\epsilon}$ has a blue arrow pointing down to $\epsilon \sim \mathcal{N}(\mu_R, \sigma_R^2) \rightarrow \text{mean, var as } R$.

1. $\lambda_i(X) = 1 \rightarrow$ Transmit all information ($Z_i = R_i$)
2. $\lambda_i(X) = 0 \rightarrow$ All information in R_i is lost ($Z_i = \epsilon$)

[Tishby et al. 2000]

Proposed Method

- Information Bottleneck for attribution:

Now, it is required to estimate how much information is contained in Z from R

$$I[R, Z] = \mathbb{E}_R[D_{KL}[P(Z|R) \| P(Z)]]$$

$p(z) = \int p(z|r)p(r)dr \rightarrow$ intractable

Variational approximation $Q(Z) = \mathcal{N}(\mu_R, \sigma_R)$ (Assumption : All dim of Z are distributed normally and independent.).

$$\begin{aligned} I[R, Z] &= \int p(R) \left(\int p(z|r) \log \frac{p(z|r)}{p(z)} dz \right) dr \\ &= \int \int p(r, z) \log \frac{p(z|r)}{p(z)} \frac{q(z)}{q(z)} dz dr \\ &= \int \int p(r, z) \log \frac{p(z|r)}{q(z)} dz dr + \int \int p(r, z) \log \frac{q(z)}{p(z)} dz dr \\ &= \int \int p(r, z) \log \frac{p(z|r)}{q(z)} dz dr + \int p(z) \left(\int p(r|z) dr \right) \log \frac{q(z)}{p(z)} dz \\ &= \mathbb{E}_R[D_{KL}[P(Z|R) \| Q(Z)]] - D_{KL}[P(Z) \| Q(Z)] \\ &\leq \mathbb{E}_R[D_{KL}[P(Z|R) \| Q(Z)]] \end{aligned}$$

[Klambauer et al. 2017]

Proposed Method

- Information Bottleneck for attribution:

$$I[R, Z] = \mathbb{E}_R[D_{KL}[P(Z|R) \| Q(Z)]] - D_{KL}[P(Z) \| Q(Z)]$$

$\mathcal{L}_I \geq I[R, Z]$

If $\mathcal{L}_I = 0$ for an area $\rightarrow$ information from this area is not necessary for the network's prediction

Their goal is keeping only the information necessary for correct classification. Thus, the mutual information should be minimal while the classification score remain high.

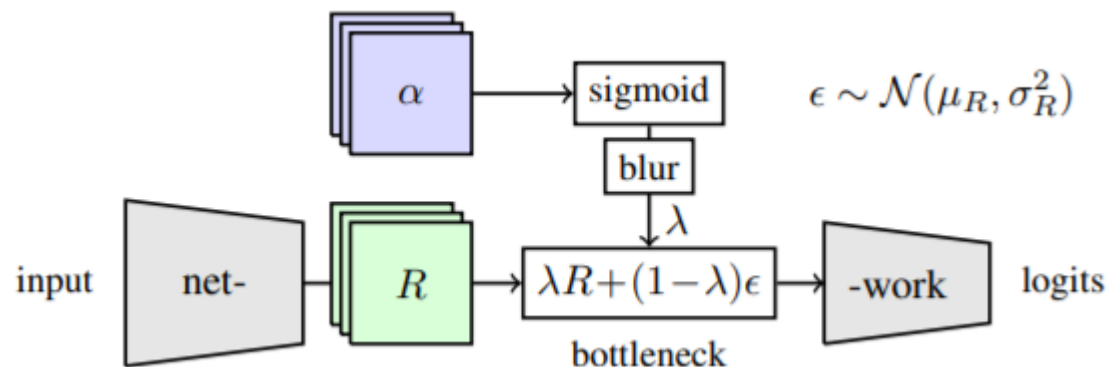
Total objective function:

$$\mathcal{L} = \mathcal{L}_{CE} + \beta \mathcal{L}_I$$

β controls the relative importance of both objectives. (e.g., for a small β , more bits of information are flowing and less for a higher β)

Proposed Method

- Per-Sample Bottleneck



Parameterization

: The bottleneck parameters λ have to be in $[0,1]$. Therefore, they parametrize $\lambda = \text{sigmoid}(\alpha)$, where $\alpha \in \mathbb{R}^d$.

Initialization

: In the beginning, they wanted retain all the information $\rightarrow$ initialize $\alpha_i = 5$, thus $\lambda \approx 0.993 \Rightarrow Z \approx R$.

At first, the bottleneck has practically no impact on the model performance. It then deviates from this starting point to suppress unimportant regions.

Optimization

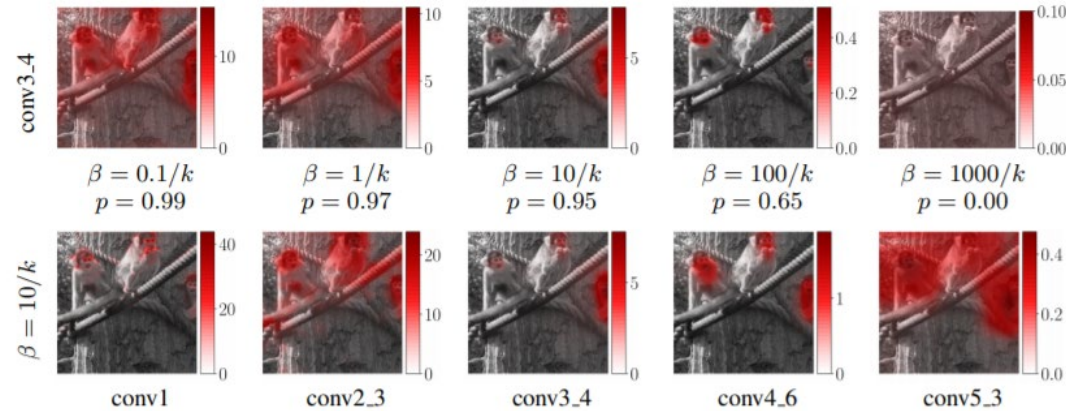
: 10 iters using Adam with $\text{lr}=1$ to fit the mask α . To stabilize the training, they copy the single sample 10 times and apply different noise to each.

Proposed Method

- Per-Sample Bottleneck

Measure of information in Z (i.e., $D_{KL}(P(Z|R) \| Q(Z))$) per dimension)

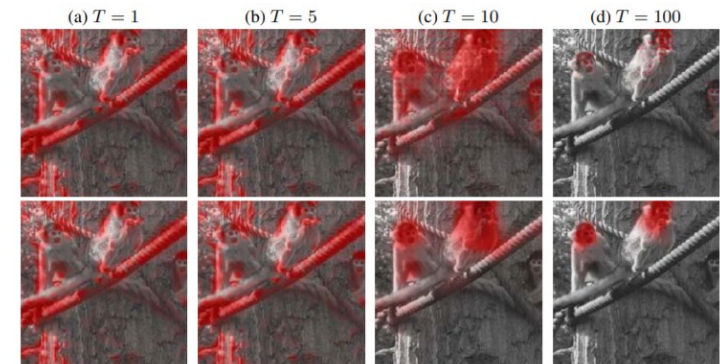
: summing over the channel axis $\rightarrow m_{[h,w]} = \sum_{i=0}^c D_{KL}(P(Z_{[i,h,w]} | R_{[i,h,w]}) \| Q(Z_{[i,h,w]}))$



Enforcing local smoothness

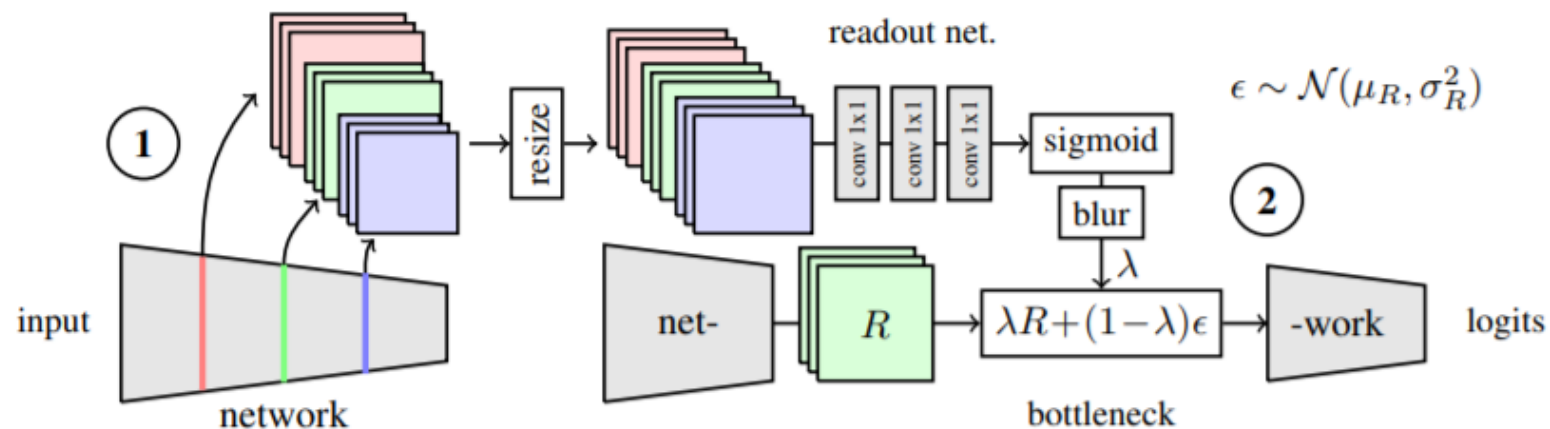
: Pooling and conv stride ignore parts of the input, causing PSB overfit to a grid structure
 $\rightarrow$ Convolve the sigmoid output with a fixed Gaussian kernel with standard deviation σ_s .

$$\lambda = \text{blur}(\sigma_s, \text{sigmoid}(\alpha))$$



Proposed Method

- Readout Bottleneck

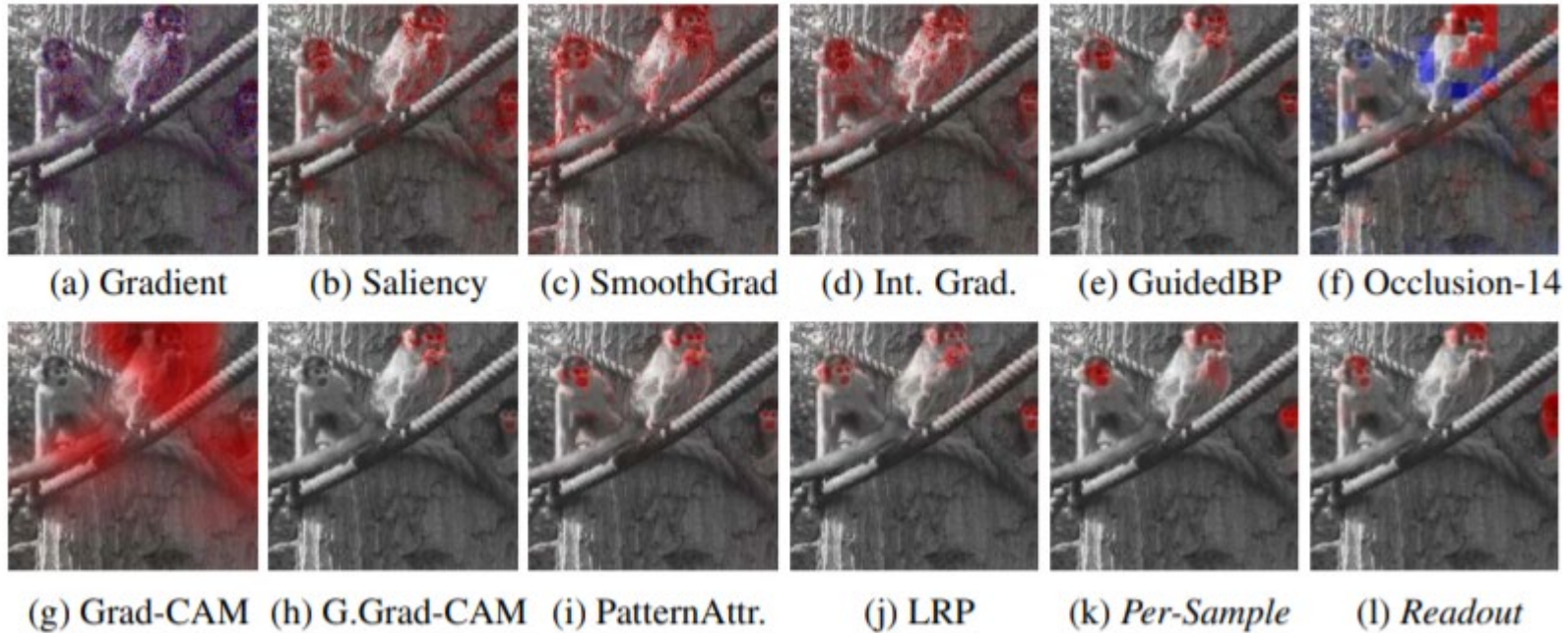


Collect feature maps from different depths and then combine them with 1X1 conv

- ① In a first forward pass, no noise is added and collect different feature maps and interpolate them bilinearly to match the spatial dimension.
- ② In a second forward pass, they insert the bottleneck layer into the network and restrict the flow of information.

Evaluation

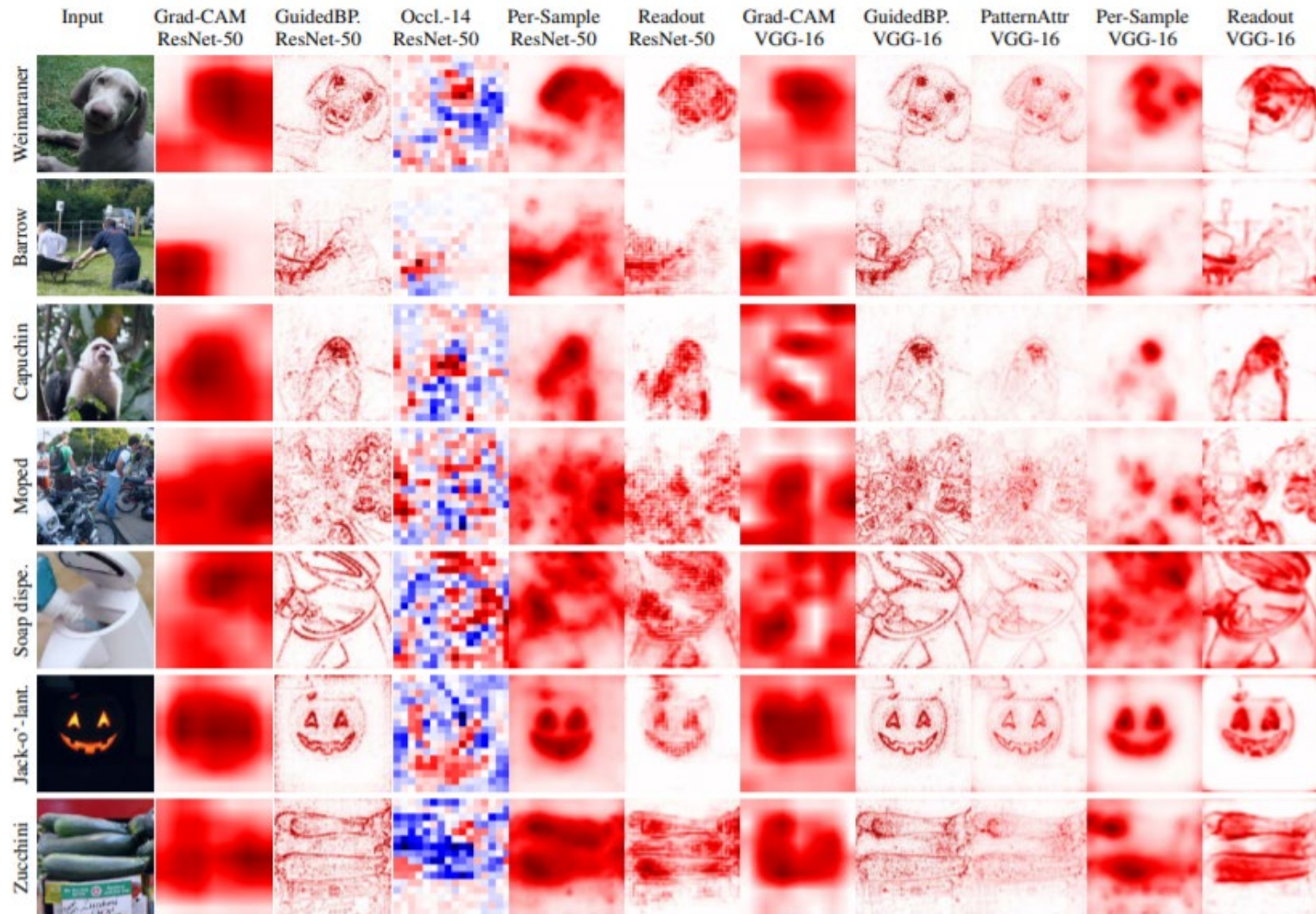
- Qualitative Assessment



Subjectively, both the PSB, RB identify areas relevant to the classification well and more specific (fewer pixels are scored high.).

Evaluation

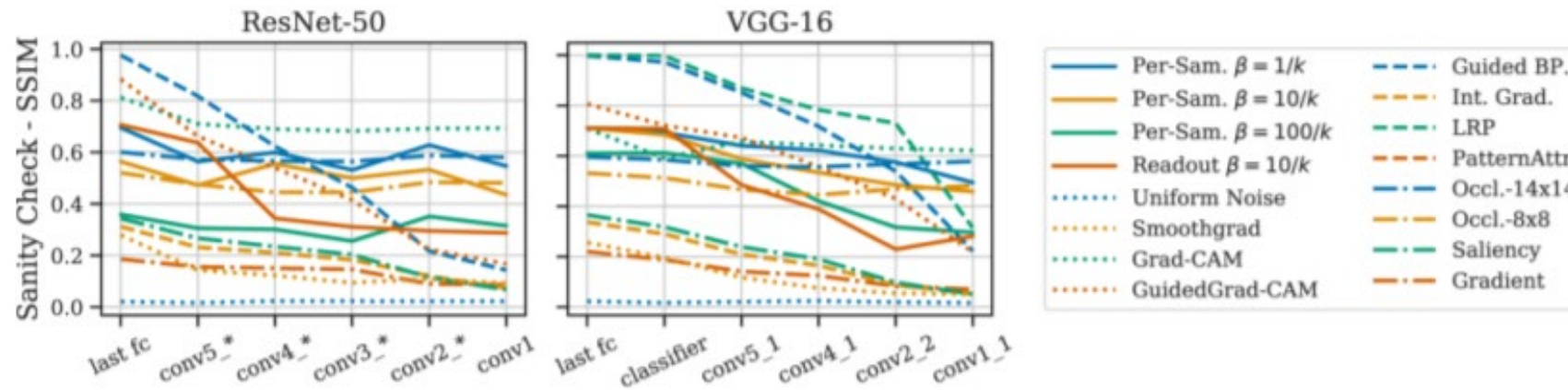
- Qualitative Assessment



Evaluation

- Sanity Check (Randomization of Model Parameters)

: Starting from the last layer, an increasing proportion of the network parameters is re-initialized until all parameters are random. The difference between original heatmap and the heatmap obtained from the randomized model is quantified using SSIM.



For their methods, the randomizing the final dense layer drops the mean SSIM by around 0.4

[Adebayo et al. 2018]

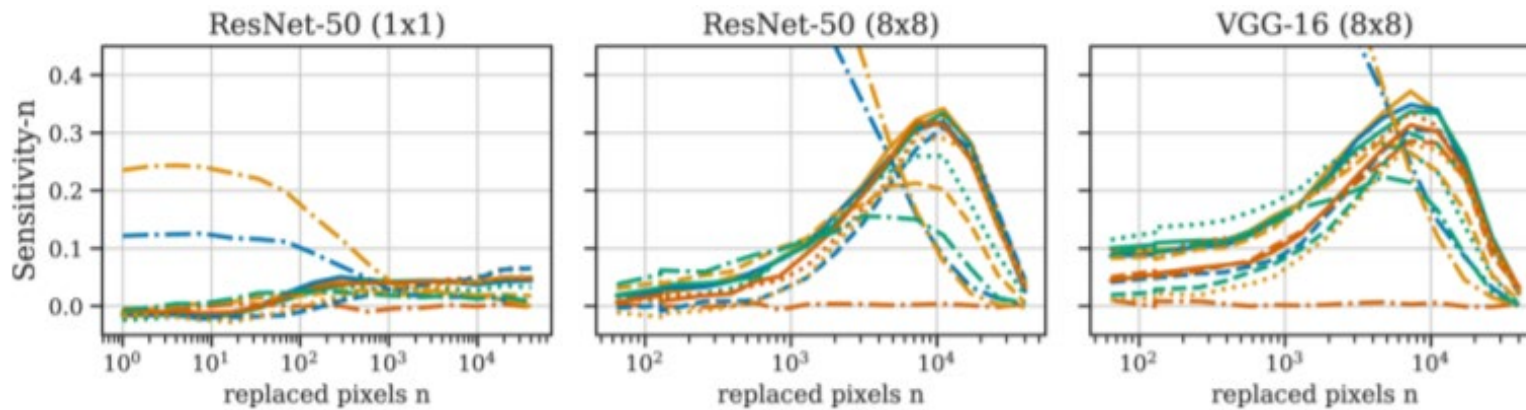
Evaluation

- Sensitivity-N

: Masks the network's input randomly and then measures how strongly the amount of attribution in the mask correlates with the drop in classifier score.

$$\text{corr} \left(\sum_{i \in T_n} R_i(x), \boxed{S_c(x)} - S_c \left(x_{\boxed{x_{T_n}=0}} \right) \right)$$

Classifier logit output for class c
The input with all pixels in T_n set to zero



PSB ($\beta = 10/k$) perform best for both models above $n = 2 \cdot 10^3$ pixels (i.e., when more than 2% of all pixels are masked).

[Anaconda et al. 2018]

Evaluation

- Localization

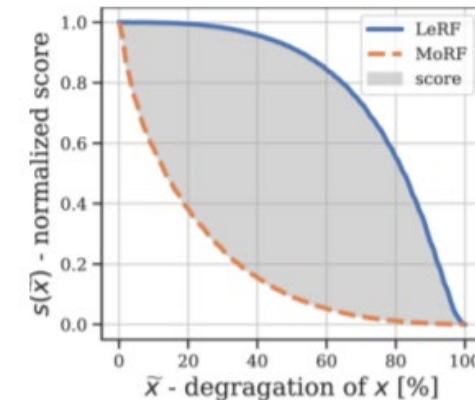
1. Bounding Box: If the bbox contains n pixels, measuring how many of n -th highest scored pixels are contained in the bounding box (Then divided by n = ratio).
2. Image Degradation: The most relevant tiles are removed first (MoRF) $\Rightarrow$ Removing tiles ranked as least relevant by the attribution method first (LeRF).

$$s(x) = \frac{p(y|x) - \boxed{b}}{\boxed{t_1} - b}$$

Mean model output on the fully degraded images.

Top-1 probability on the original samples

Model & Evaluation	ResNet-50 deg.		VGG-16 deg.		ResNet	VGG
	8x8	14x14	8x8	14x14	bbox	bbox
Random	0.000	0.000	0.000	0.000	0.167	0.167
Occlusion-8x8	0.162	0.130	0.267	0.258	0.296	0.312
Occlusion-14x14	0.228	0.231	0.402	0.404	0.341	0.358
Gradient	0.002	0.005	0.001	0.005	0.259	0.276
Saliency	0.287	0.305	0.326	0.362	0.363	0.393
GuidedBP	0.491	0.515	0.460	0.493	0.388	0.373
PatternAttribution	—	—	0.440	0.457	—	0.404
LRP $\alpha=1, \beta=0$	—	—	0.471	0.486	—	0.397
LRP $\alpha=0, \beta=1, \epsilon=5$	—	—	0.462	0.467	—	0.441
Int. Grad.	0.401	0.424	0.420	0.453	0.372	0.396
SmoothGrad	0.485	0.502	0.438	0.455	0.439	0.399
Grad-CAM	0.536	0.541	0.510	0.517	0.465	0.399
GuidedGrad-CAM	0.565	0.577	0.555	0.576	0.468	0.419
IBA Per-Sample $\beta=1/k$	0.573	0.573	0.581	0.583	0.606	0.566
IBA Per-Sample $\beta=10/k$	0.572	0.571	0.582	0.585	0.620	0.593
IBA Per-Sample $\beta=100/k$	0.534	0.535	0.542	0.545	0.574	0.568
IBA Readout $\beta=10/k$	0.536	0.536	0.490	0.536	0.484	0.437



Both LeRF, MoRF degradation yield curves measuring different qualities of the attribution method.
 $\rightarrow$ calculate the integral between the two curves

Thank You Q&A

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